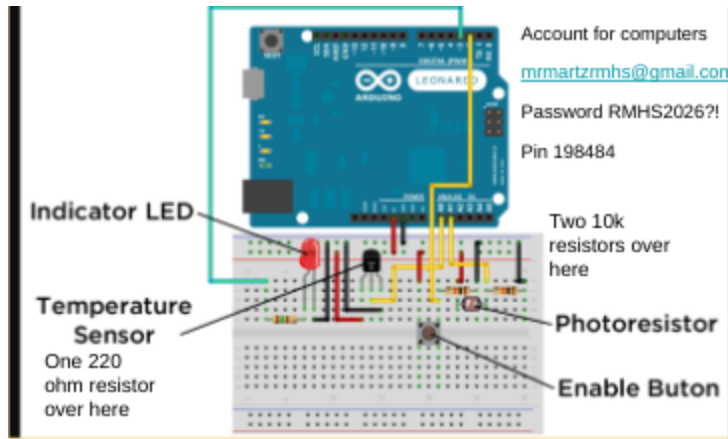




Arduino is like a microprocessor of data. In this product, Arduino plays an important role because it gets the value of the temperature from the temperature sensor, and light level from the photoresistor, then it prints the temperature first, then comma, and finally light level in a document, after that, it uses a new line character to separate the readings. Without it, no data about temperature and light level can be recorded in any way.

When considering the role of the arduino, it is heavily involved in technology and engineering aspects of STEM practices. Using MATLAB as a program to help solve this problem requires knowledge and experience in coding. But when taking in consideration the reason to gather light intensity, humidity, and temperature data, knowledge and application of science is required. Largely, it was to measure how these three components could affect the growth rate of the plants in a greenhouse. When gathering these three values, the science part of STEM is required to understand what they are used for.

We set the arduino to notify us when humidity reaches above 95%. Humidity when too high can grow mold and cause the plant's roots to rot, making it an issue for the plant to get nutrients while also making it vulnerable to diseases. Measuring light intensity is also important because the rate of photosynthesis depends on how much light the leaves are able to absorb within a period of time, which then get turned into chemical energy to take part of the photosynthesis process. Lastly, increasing temperature generally increases the rate of growth for plants as compared to lower temperatures. Extreme heat or cold can lead to problems with photosynthesis and respiration, which would stunt plant growth too.

## Element E: Code Explanation

|   |                                                                                    |                                                                                                                                                                                                                               |
|---|------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1 | <code>clf</code>                                                                   | clf means clear figure which clears the line plot.                                                                                                                                                                            |
| 2 | <code>M = readmatrix('ead_2.csv')</code>                                           | It reads all of the data in “ead_2.csv”.<br><br>“Readmatrix” creates an array by reading column-oriented data from a file. The readmatrix function performs automatic detection of import parameters for your file.           |
| 3 | <code>figure (1)</code>                                                            | figure creates a new figure window using default property values. So it uses the “ead_2.csv” data for figure 1                                                                                                                |
| 4 | <code>plot(M(:,1),M(:,2));</code>                                                  | It sets points on the graph using matrices for figure 1                                                                                                                                                                       |
| 5 | <code>axis ([0 150 300 400])</code>                                                | It determines how big the graph is going to be for figure 1                                                                                                                                                                   |
| 6 | <code>figure (2)</code>                                                            | figure creates a new figure window using default property values. So it uses the “ead_2.csv” data for figure 2                                                                                                                |
| 7 | <code>plot(M(:,1),M(:,3));</code>                                                  | It sets points on the graph using matrices for figure 2                                                                                                                                                                       |
| 8 | <code>axis ([0 150 800 1020])</code>                                               | It determines how big the graph is going to be for figure 2                                                                                                                                                                   |
| 9 | <code>% column 1 = event column 2<br/>= temp column 3 = light<br/>intensity</code> | It sorts the data into columns, <b>making column 1 be the event</b> in which it's happening (number order), <b>column 2 records the temperature data for each row</b> , and <b>column 3 records the light intensity data.</b> |

All of this code allows the data in “ead\_2.csv” to be converted into a graph with all the data points within the set axis' for both independent figures.